CPU vs OpenCL Image Processing Benchmark

ROS 2 Jazzy • OpenCV Gaussian Blur • NVIDIA GeForce RTX 2050

1. Objective

This benchmark compares CPU and OpenCL execution of an OpenCV 31×31 Gaussian Blur using synthetic BGR images at three resolutions: 640×480, 1280×720, and 3840×2160. Each resolution/mode combination was measured across three runs.

2. Test environment

Component	Configuration
Operating system	Ubuntu 24.04
ROS 2	Jazzy
Processor	Intel Core i5-11320H (4 cores / 8 threads)
GPU	NVIDIA GeForce RTX 2050 (4 GB VRAM)
OpenCV	4.6.0 with OpenCL support
OpenCL device	NVIDIA GeForce RTX 2050
Operation	31×31 Gaussian Blur

3. Mean latency results

Resolution	CPU mean (ms)	OpenCL mean (ms)	Comparison
640×480	4.326	3.558	OpenCL 17.7% lower
1280×720	8.373	6.777	OpenCL 19.1% lower
3840×2160	39.666	48.525	OpenCL 22.3% higher

4. Interpretation

OpenCL produced lower mean latency at 640×480 and 1280×720 in these tests. At 4K, CPU processing was faster. A likely contributing factor is the overhead of transferring image data between host memory and GPU memory, in addition to synchronization. The results show that GPU acceleration is workload- and resolution-dependent; it is not automatically faster for every image size.

5. Methodology and limitations

• The input is a deterministic synthetic BGR image generated by the image publisher.

• The measured operation is a 31×31 OpenCV Gaussian Blur.

• CPU timing measures the GaussianBlur call. OpenCL timing includes Mat-to-UMat upload, blur, UMat-to-Mat download, and synchronization. The timing boundaries are therefore not identical.

• At 640×480 and 1280×720, each run used 20 warm-up frames and 100 measured frames.

• At 4K, CPU runs used 20 warm-up and 100 measured frames; OpenCL runs used 5 warm-up and 30 measured frames. The smaller OpenCL sample count should be considered when interpreting the 4K comparison.

• The reported FPS is calculated as 1000 divided by mean processing latency. It is not a measurement of end-to-end ROS image-stream throughput.

• OpenCL device selection was confirmed, but this alone does not establish sustained GPU utilization or rule out internal fallback behavior.

6. Conclusion

For this implementation and test environment, OpenCL reduced mean processing latency at the two lower resolutions, while CPU processing was faster at 4K. Future work should include repeated runs with equal sample counts, additional image-processing operations, and separate measurements of transfer time, kernel time, and end-to-end pipeline throughput.

Raw per-run measurements are provided separately in results/benchmark_results.csv.